

2.1 The Structure of the Internet

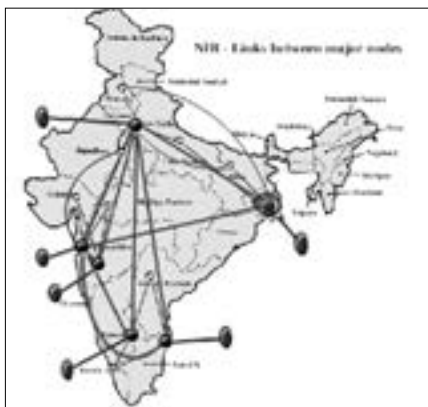
2.1.1 Webs In General

Spiders made webs before we did, you know. At first, a spider makes a bridge line—the first thread that connects two solid points, like two branches or walls or trees. He then reinforces this thread with a few more threads. This primary line is what holds the entire web together. He then proceeds to make a Y-shaped radial—a framework—and then the spokes that constitute a web.

If you thought it was just some bright copywriter's idea to call the Internet a “web,” think again. There are some remarkable similarities between how spiders make their webs and how the Internet takes shape. The primary resemblance is what is known as the Internet “backbone.”

2.1.2 The Central—But Not Really Nervous—System

Akin to the spider's bridge line is what is called a “backbone,” a super-fast network that goes all across the world, usually from one metropolitan area to another. Mega- ISPs run these lines that can transport data at approximately 45 MBps (on T3 lines) and are linked at specified interconnection points (called “national access points” and sometimes known as “regional nerve centres”). All regional networks are connected to each other by high-speed backbones, which are basically connections that can transport data at very high speeds. When data is sent from one regional network to another, it is first sent to the above-mentioned NAP which in turn routes it to a backbone. The ISP that connects your computer to the Internet is a local one that uses routers to send and receive data via this backbone.



The BSNL National Internet Backbone (NIB)